TRACK 1 | MERCHANT GROWTH AI

DukaanSaathi AI

दुकान साथी — हर दुकान का अपना AI साथी

The AI-Powered Business Copilot That Transforms Every Paytm Merchant's Transaction Data Into Actionable Growth Strategies

Voice-First AI | Demand Forecasting | Smart Inventory | Customer Retention | Agentic Actions

Paytm Build for India AI Hackathon — Delhi Edition 2026

Team Lead: Tushar Bhardwaj | NIT Kurukshetra

PROBLEM STATEMENT

30M+ Paytm Merchants Sit on a Data Goldmine They Can't See

Small merchants generate rich transaction data daily — but have zero tools to turn it into business decisions.

Zero Data Literacy

85%+ merchants can't interpret their own sales patterns or seasonal trends

Blind Inventory

Rs.32,000 Cr annual waste from overstocking in Indian retail alone

No Demand Forecasting

Merchants rely on gut feel, miss festival & weather driven demand spikes

Customer Churn Invisible

No way to know which regular customers are slipping away

No Growth Roadmap

No structured path from Rs.10K/month to Rs.1L/month — merchants plateau

Pricing Guesswork

No data-driven approach to pricing, competitor awareness, or discounts

CORE INSIGHT: Paytm has the richest merchant transaction data in India. The problem isn't data — it's that this goldmine is invisible to merchants.

PROPOSED SOLUTION

DukaanSaathi AI: 4-Module AI Copilot System

A multi-agent AI system with 4 specialized intelligence modules, unified by a Hindi-first conversational voice interface.

1 Demand Oracle AI

"Kal kya bikega?"

Hyper-local demand forecasting with weather, festival & event awareness. Predicts what will sell before the merchant asks.

XGBoost + LSTM Hybrid

2 Smart Inventory AI

"Kya stock karna hai?"

Camera-based shelf scanning + knowledge graph recommendations. Snap a photo — AI tells you what to reorder & when.

YOLO + Knowledge Graphs

3 Customer Retention AI

"Kaun chhod raha hai?"

RFM segmentation + churn prediction. Identifies at-risk customers & auto-generates win-back cashback offers.

K-Means + Survival Analysis

4 Growth Path Navigator

"Aage kaise badhein?"

Gamified milestone roadmap with peer benchmarking. MerchantMind Score (0-900) for credit & growth tracking.

Collaborative Filtering

ALL MODULES ACCESSIBLE VIA: Voice (Hindi/Regional) | Text Chat | Camera Scan | Push Notifications

WHAT MAKES US DIFFERENT — 3 HERO FEATURES

Voice-First AI Copilot

Sarvam AI Speech-to-Speech

Merchant SPEAKS in Hindi or any regional language. AI SPEAKS back with actionable business advice.

No typing. No dashboards.
No digital literacy needed.

"Bhaiya, aaj kya bikega?"
AI: "Tushar ji, aaj barish ka forecast hai. Chai patti aur pakoda ingredients rakhiye — pichli barish mein aapki bikri 35% badhi thi."

Powered by Sarvam AI (Official Partner)

Snap-to-Stock Scanning

YOLO + Gemini Vision

Merchant takes a PHOTO of their shop shelf with phone camera.

AI instantly identifies products, estimates stock levels, and cross-references with demand forecast to generate alerts:

Maggi — 20% stock LEFT
High demand predicted (weekend)
REORDER 3 cartons by tomorrow

Kurkure — OUT OF STOCK
You lost Rs.800 last time
this went empty

Zero Friction Inventory — No Manual Entry

Agentic Auto-Actions

LangChain ReAct + Tool Use

AI doesn't just SUGGEST.
It ACTS with one-tap approval:

AUTO-REORDER: Detects low stock, sends WhatsApp order to saved supplier contact

WIN-BACK OFFERS: Detects churning customer, sends Rs.20 cashback via Paytm

CASH FLOW ALERT: Predicts shortfall next week, suggests Paytm Business Loan

Human-in-the-Loop: Merchant Always Approves

ADDITIONAL INTELLIGENCE LAYER

3 More Capabilities That Complete the AI Copilot

Beyond forecasting & inventory, DukaanSaathi AI adds financial intelligence, credit scoring, and competitive awareness.

4 Predictive Cash Flow Engine

"Will I have enough money next week?"

Predicts cash crunches BEFORE they happen by analyzing:

- Sales trend forecasts
- Recurring expenses (rent, EMIs)
- Supplier payment cycles

When shortfall is predicted:
- Alerts merchant 7 days early
- Suggests Paytm Business Loan (pre-approved, one-tap apply)

No merchant in India has access to cash flow forecasting today.

Monte Carlo Simulation + Recurrence Detection

5 MerchantMind Score (0 - 900)

AI-Generated Merchant Credit Score

5-dimension health score:

- Revenue Consistency (25%)
- Customer Loyalty (20%)
- Transaction Diversity (15%)
- Growth Velocity (25%)
- Digital Maturity (15%)

750+ Score unlocks:
- Lower interest Paytm Loans
- 'Trusted Merchant' badge
- Priority support

Creates a NEW financial product.

XGBoost Scoring + SHAP Explainability

6 Hyperlocal Competitor Radar

"What's working for shops near you?"

Anonymized, aggregated intel from nearby same-category shops:

- Trending products in area
- Top 10% peer benchmarks
- New opportunity alerts
- Competitive threat warnings

Example:
'3 kiranas near you started selling organic atta and saw 34% higher revenue this month'

Privacy-safe aggregated data only.

H3 Geospatial + Z-Score Anomaly Detection

4 Core Modules + 6 Enhancements = Complete AI Business Copilot for 30M+ Paytm Merchants

TECHNOLOGY / TECH STACK

End-to-End AI Architecture

INTERFACE LAYER	**Sarvam AI** Indic ASR + TTS Hindi-first Voice	**YOLO + Gemini Vision** Camera Shelf Scanning	**Next.js Dashboard** Interactive Web Visualization	**WhatsApp API** Business Bot Integration
AI ORCHESTRATION	**LangChain** Multi-Agent Orchestration	**Google Gemini 2.5** Reasoning & Generation	**ReAct Agents** Agentic Tool-Use Actions	**ChromaDB** Vector Store for Semantic Search
ML ENGINE	**XGBoost + LSTM** Demand Forecasting	**K-Means + RFM** Customer Segmentation	**NetworkX** Product Knowledge Graphs	**SHAP** Explainable AI Score Attribution
INFRASTRUCTURE	**FastAPI** REST + SSE Streaming Backend	**PostgreSQL** TimescaleDB for Time-Series	**Docker** Containerized Deployment	**Azure Cloud** Production Hosting

STRATEGIC: Sarvam AI (Official Hackathon AI Partner) used for Hindi-first voice interface — Indic ASR + TTS models

UNIQUE SELLING PROPOSITION

Why No Competitor Can Match DukaanSaathi AI

Hindi-First Voice AI

Accessibility

Not a dashboard merchants won't use. A voice assistant that speaks their language — literally.

Hyper-Local Context

Intelligence

Combines weather + festivals + local events + neighborhood trends at individual merchant level.

Camera Shelf Scanning

Innovation

Snap a photo of your shelf — AI tells you what's low, what to reorder. Zero manual data entry.

Agentic Actions

Automation

AI doesn't just suggest — it auto-orders from suppliers, sends customer offers, with one-tap approval.

MerchantMind Score

Fintech

AI-generated credit score (0-900) unlocks Paytm Business Loans at better rates. New financial product.

Paytm Ecosystem Synergy

Business Fit

Naturally drives adoption of Paytm Postpaid, Lending, Insurance, Ads — increasing Paytm's own revenue.

vs. Traditional POS Analytics: No forecasting, no voice, no camera. | vs. Generic AI Chatbots: No merchant context, no agentic actions, no Paytm integration.

IMPACT & BENEFITS

Triple Impact: Merchants + Paytm + India's Economy

FOR MERCHANTS

+15-25%
Revenue Growth
Demand-aligned stocking + customer retention

-30%
Inventory Waste
Smart reorder points + dead stock alerts

+20%
Customer Retention
RFM-driven win-back campaigns

5+ hrs/week
Time Saved
AI automates all business analysis

FOR PAYTM

+15-25%
Merchant GMV Growth

-20%
Merchant Churn

3x
Product Cross-sell

Data Moat
Competitive Differentiation

FOR INDIA'S ECONOMY

Financial Inclusion — AI business intelligence for Tier 2/3/4 merchants who can't afford consultants

Food Waste Reduction — Better demand forecasting for perishable goods across millions of kiranas

Digital Adoption — AI value proposition drives UPI & digital payments in underserved segments

BUSINESS MODEL

5 Revenue Streams for Paytm

Freemium SaaS

Free tier drives adoption
Pro Rs.199/mo | Premium Rs.499/mo

Free: 5 nudges/week, basic insights
Pro: Unlimited + demand forecasting
Premium: Full AI copilot + WhatsApp

GMV Revenue Uplift

Higher merchant sales =
higher MDR revenue for Paytm

Growing merchants transact more on Paytm. Every Rs.1 GMV increase generates direct commission.

Financial Product Cross-sell

AI recommends Paytm Loans, Insurance, Postpaid

Cash flow prediction drives loan adoption. MerchantMind Score unlocks better rates.

Paytm Ads Platform

Brands pay to be recommended ethically

FMCG brands sponsor inventory suggestions (disclosed). E.g., "Try stocking Brand X — similar stores"

Data Intelligence

Anonymized demand intelligence for brands

Aggregated, privacy-safe demand trends sold to FMCG/CPG brands for supply chain planning.

REVENUE PROJECTIONS (CONSERVATIVE)

Metric	Year 1	Year 2	Year 3
Pro Subscribers (2% of free)	100K	300K	500K
Subscription Revenue	**Rs.39 Cr/yr**	**Rs.117 Cr/yr**	**Rs.195 Cr/yr**
Incremental GMV Revenue	**Rs.100+ Cr/yr**	**Rs.400+ Cr/yr**	**Rs.800+ Cr/yr**

DukaanSaathi AI

हर दुकान का अपना AI साथी

Making AI-powered business intelligence accessible to every merchant in India

TEAM

Tushar Bhardwaj

NIT Kurukshetra | AI/ML Researcher | Full-Stack AI Builder

20+ Production AI Projects | 1500+ DSA Problems

Paytm Build for India AI Hackathon 2026 | Track 1: Merchant Growth AI | Let's Build for India